7.5 Properties of Trapezoids and Kites

Name _____ Period ____ Date _____

Geometry 7.5 · Student sheet

I can…

- find the angle measures of an isosceles trapezoid.
- use the Trapezoid Midsegment Theorem to find lengths.
- find the angle measures of a kite and name a quadrilateral as specifically as possible.

Vocabulary · fill in the missing word

Trapezoid — a _____ with exactly one pair of parallel sides (the bases); the other sides are the legs

Isosceles trapezoid — a trapezoid with _____ legs

Base angles — a pair of angles that _____ a base

Midsegment of a trapezoid — the segment that joins the _____ of the legs

Kite — a _____ with two pairs of consecutive congruent sides, but opposite sides not congruent

Part 1 · Fold and cut

Fold a sheet in half and cut a shape that unfolds into an **isosceles trapezoid**. Then fold another sheet and cut a **kite**. Measure.

Isosceles trapezoid	Measure
Base angles (bottom)	
Base angles (top)	
Diagonals	

Kite	Measure
The pair of angles that match	
The other two angles	
Angle where the diagonals cross	

Two things that are always true about each shape:

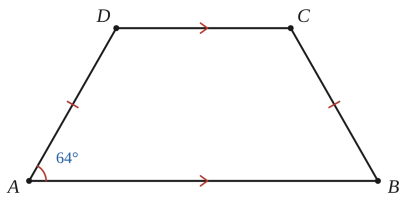
Part 2 · Notes and examples (fill in each note, then work the example)

Trapezoid: A **trapezoid** has exactly one pair of _____ sides (the bases).

Isosceles trapezoid: In an isosceles trapezoid, each pair of base angles is _____, and the diagonals are _____.

Example 1

ABCD is an isosceles trapezoid with bases $\overline{AB}$ and $\overline{CD}$, and $m\angle A = 64°$. Find $m\angle B$, $m\angle C$, and $m\angle D$.



Trapezoid Midsegment Theorem: the midsegment is _____ to the bases, and its length is _____ the sum of the bases.

Example 2

a. The bases of a trapezoid are 12 and 20. Find the length of the midsegment.

b. The midsegment of a trapezoid is 15 and one base is 11. Find the other base.

Kite: In a kite, the diagonals are _____, and exactly one pair of opposite angles is _____.

Example 3

In kite EFGH, the angles between the congruent sides measure 70° and 50°. Find the measure of each of the other two angles.

Example 4

Show that A(−2, 0), B(0, 3), C(4, 3), and D(6, 0) form a trapezoid. Is it isosceles?

Part 3 · Partner practice

Partner 1

The midsegment of a trapezoid is 14. The bases are $x + 4$ and $3x$. Find x and both bases.

Partner 2

In a kite, one of the congruent opposite angles measures $105°$ and one of the other angles measures $85°$. Find the fourth angle.